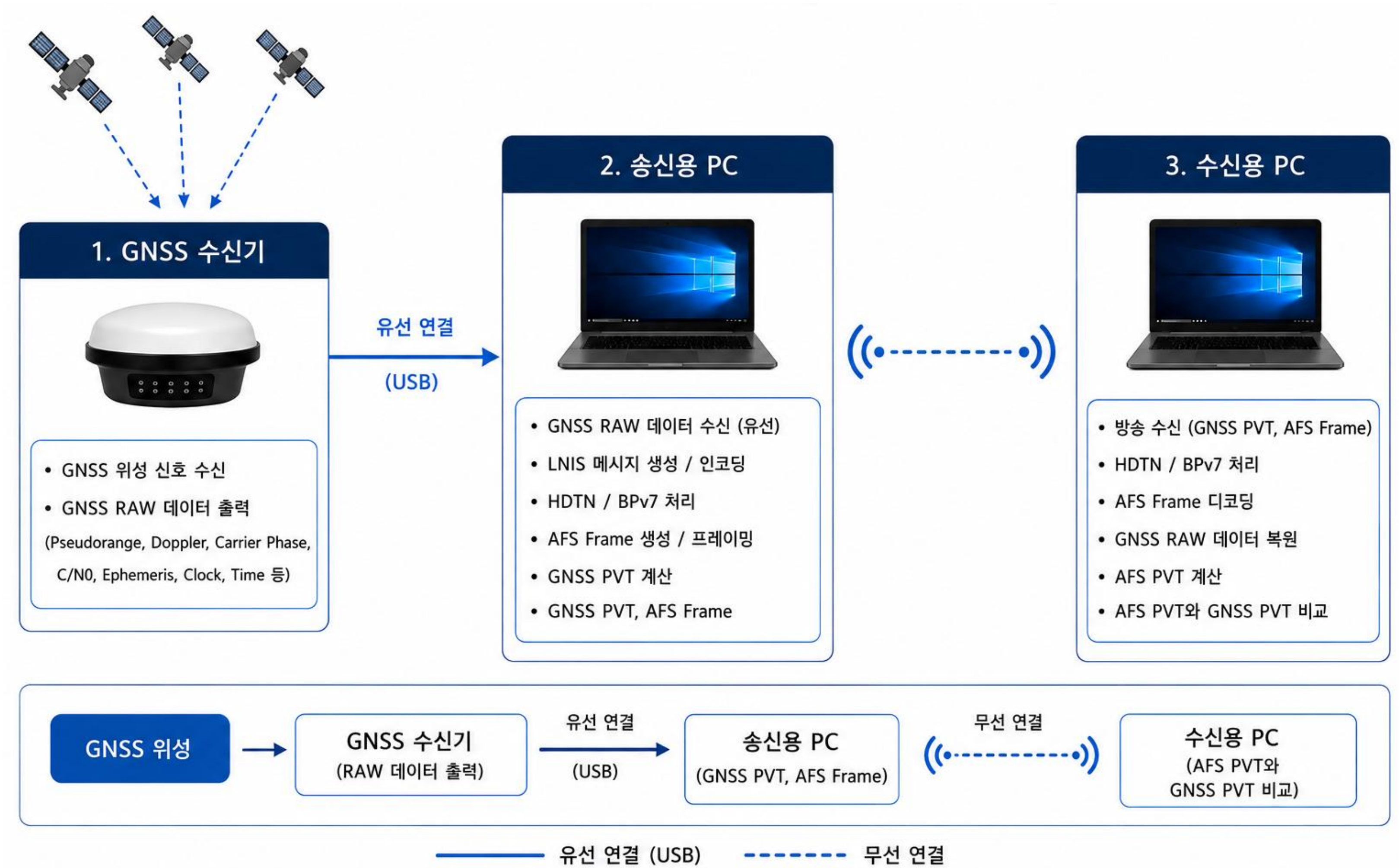
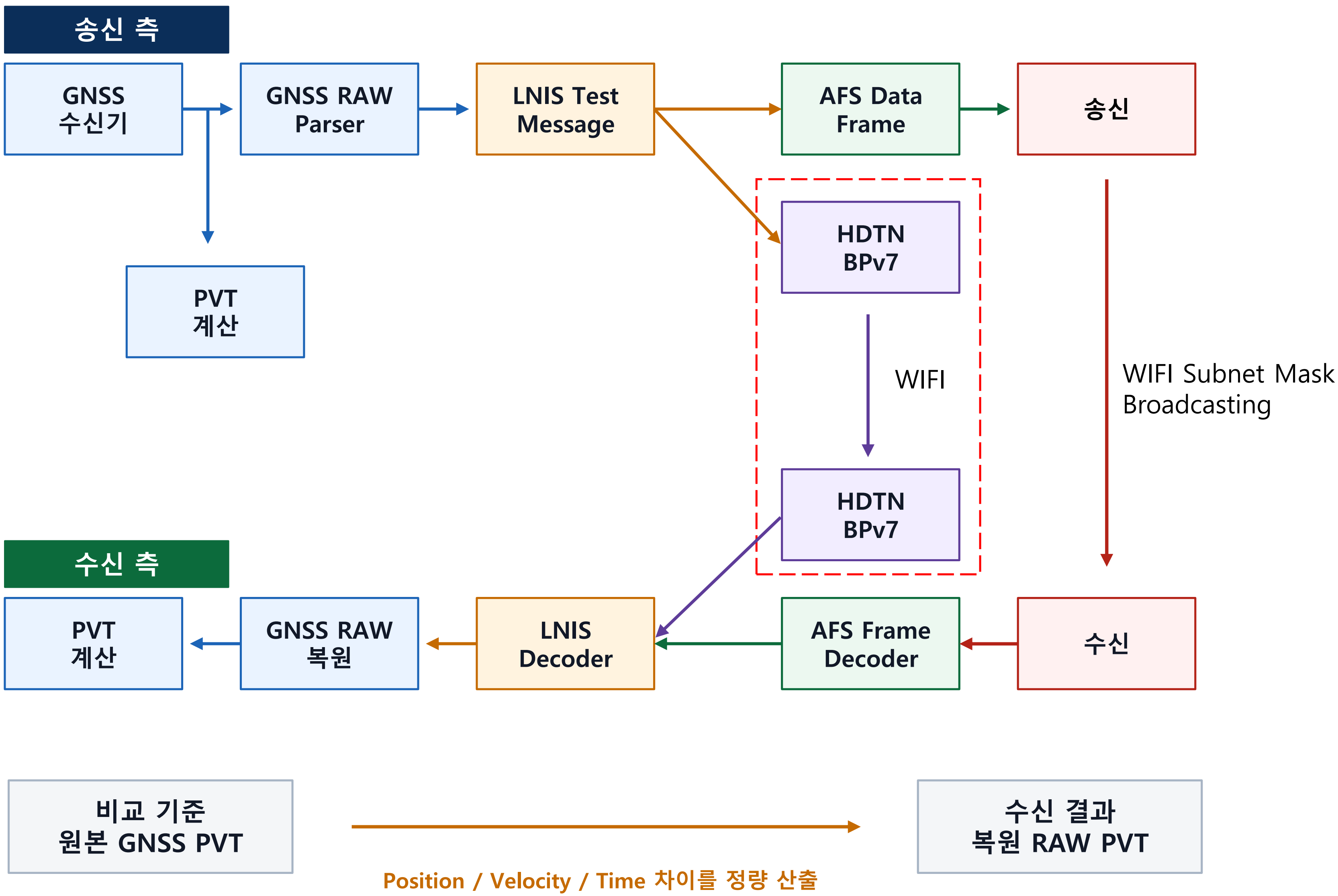
1. 시스템 구성도



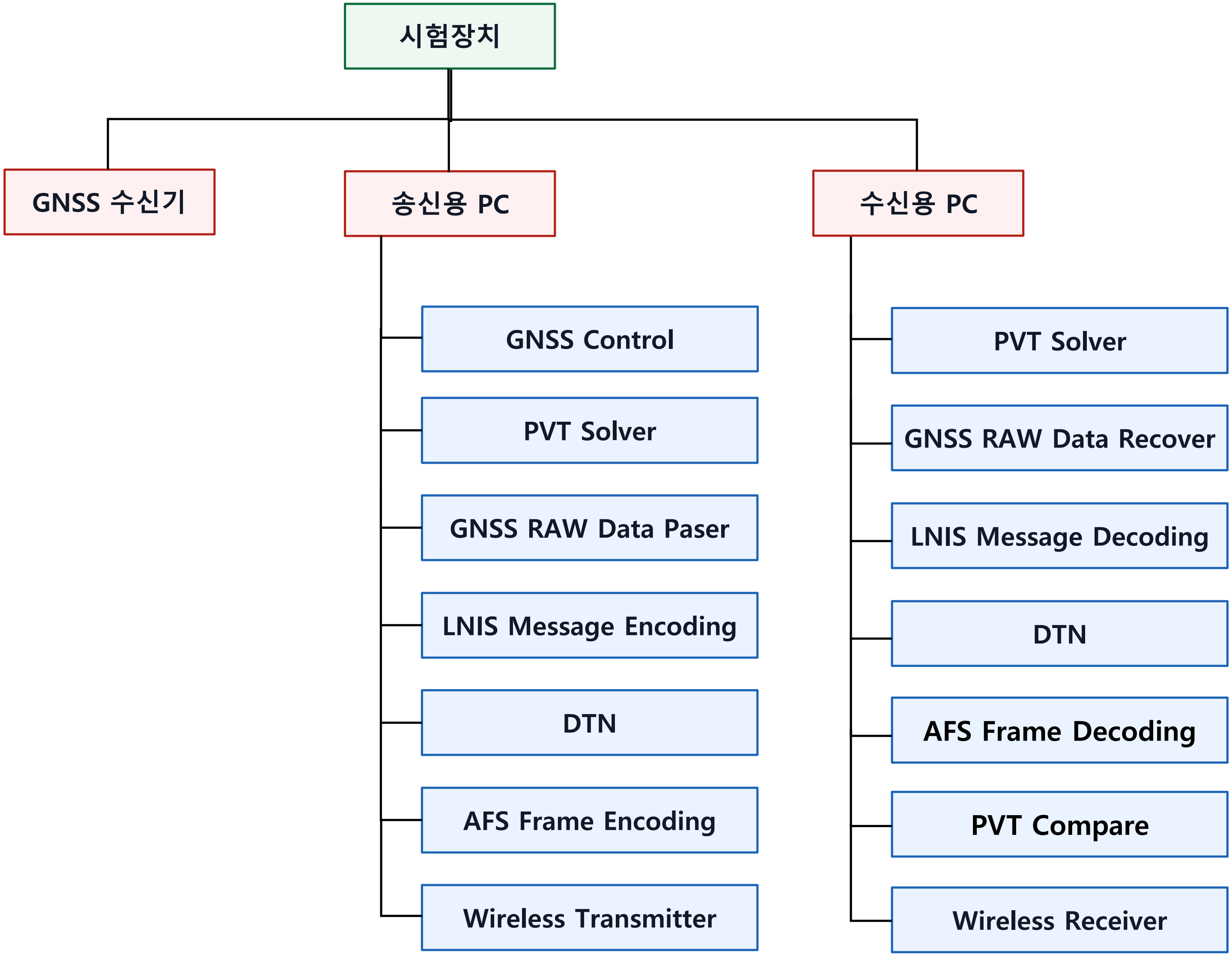
1. GNSS 수신기
• GNSS 위성 신호 수신
• GNSS RAW 데이터 출력
(Pseudorange, Doppler, Carrier Phase, C/N0, Ephemeris, Clock, Time 등)
유선 연결
(USB)
2. 송신용 PC
• GNSS RAW 데이터 수신 (유선)
• LNIS 메시지 생성 / 인코딩
• HDTN / BPv7 처리
• AFS Frame 생성 / 프레이밍
• GNSS PVT 계산
• GNSS PVT, AFS Frame
3. 수신용 PC
• 방송 수신 (GNSS PVT, AFS Frame)
• HDTN / BPv7 처리
• AFS Frame 디코딩
• GNSS RAW 데이터 복원
• AFS PVT 계산
• AFS PVT와 GNSS PVT 비교
GNSS 위성
GNSS 수신기
(RAW 데이터 출력)
유선 연결
(USB)
송신용 PC
(GNSS PVT, AFS Frame)
무선 연결
수신용 PC
(AFS PVT와
GNSS PVT 비교)
유선 연결 (USB)
무선 연결

2. 전체 구성 블럭도



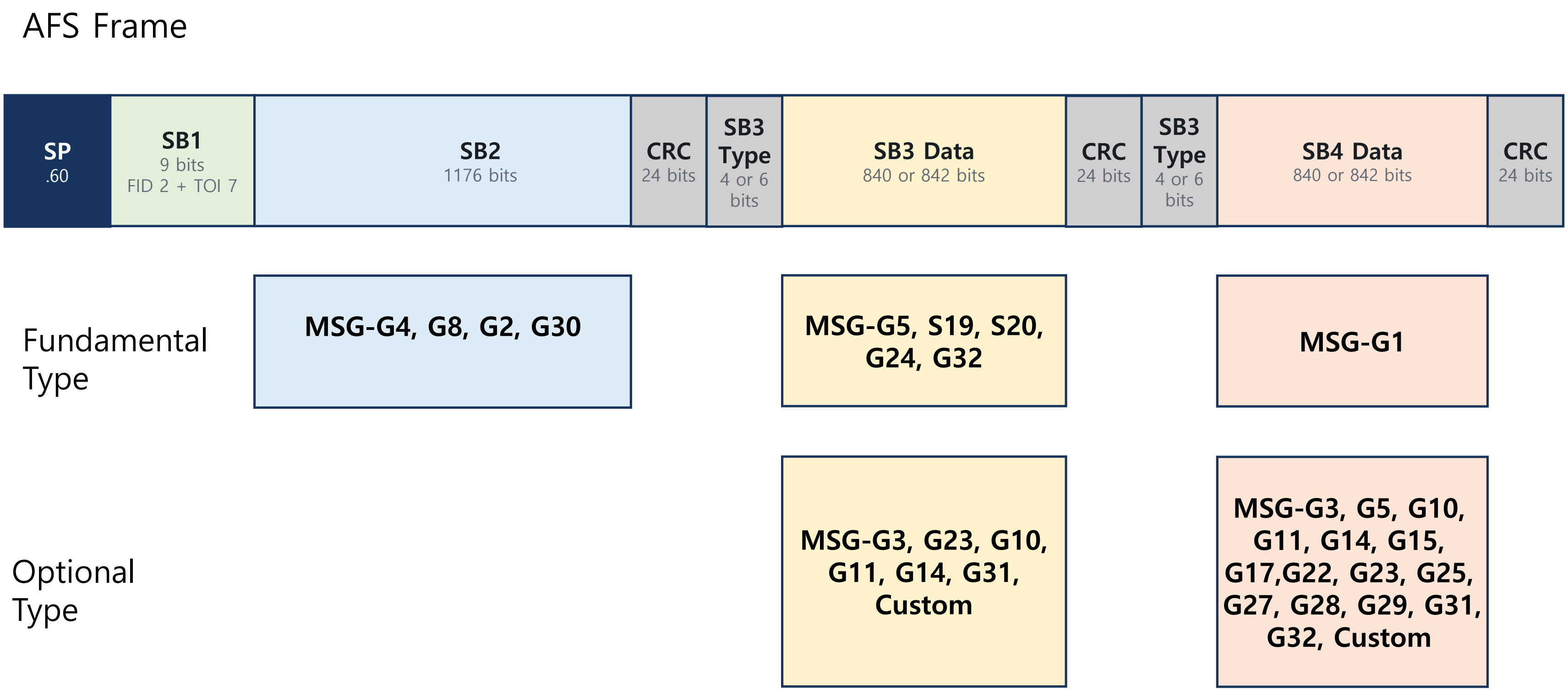
송신 측
GNSS 수신기
GNSS RAW Parser
LNIS Test Message
AFS Data Frame
송신
PVT 계산
HDTN BPv7
WIFI
HDTN BPv7
WIFI Subnet Mask Broadcasting
수신 측
PVT 계산
GNSS RAW 복원
LNIS Decoder
AFS Frame Decoder
수신
비교 기준 원본 GNSS PVT
수신 결과 복원 RAW PVT
Position / Velocity / Time 차이를 정량 산출

3. 시험장치 구성



시험장치
GNSS 수신기
송신용 PC
수신용 PC
GNSS Control
PVT Solver
GNSS RAW Data Paser
LNIS Message Encoding
DTN
AFS Frame Encoding
Wireless Transmitter
PVT Solver
GNSS RAW Data Recover
LNIS Message Decoding
DTN
AFS Frame Decoding
PVT Compare
Wireless Receiver

4. 데이터 캡슐화 관계



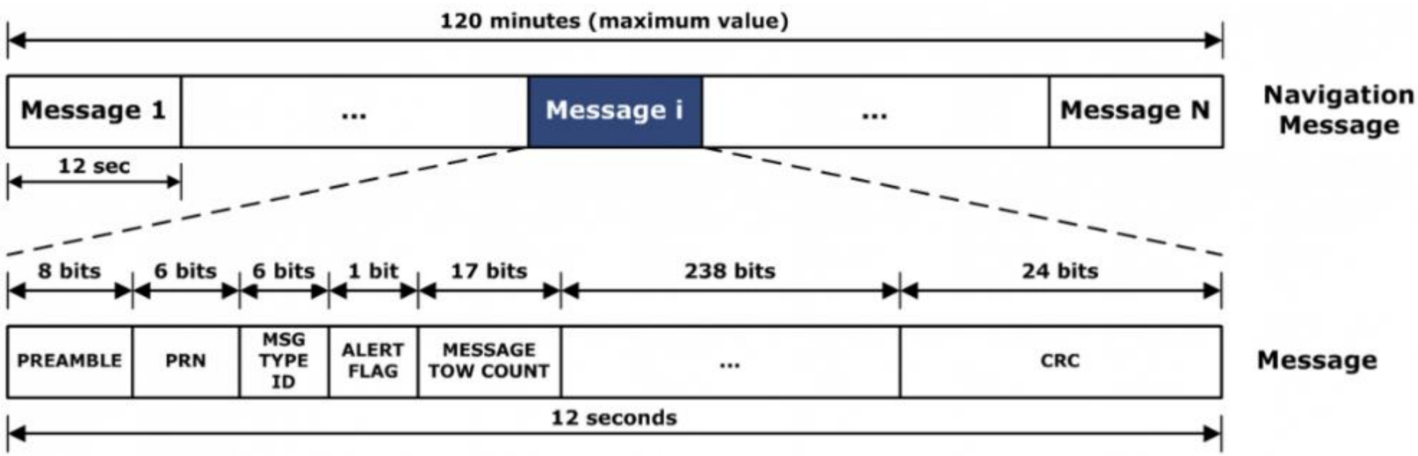
- SB3, SB4의 데이터는 프레임마다 반드시 포함 될 필요는 없음
- 데이터가 많으면 여러 프레임에 걸쳐 나눠 전송 가능
- Data Type에 대한 bit단위 내용은 정의 되어있지 않음

6. LNIS Message → AFS Subframe Allocation

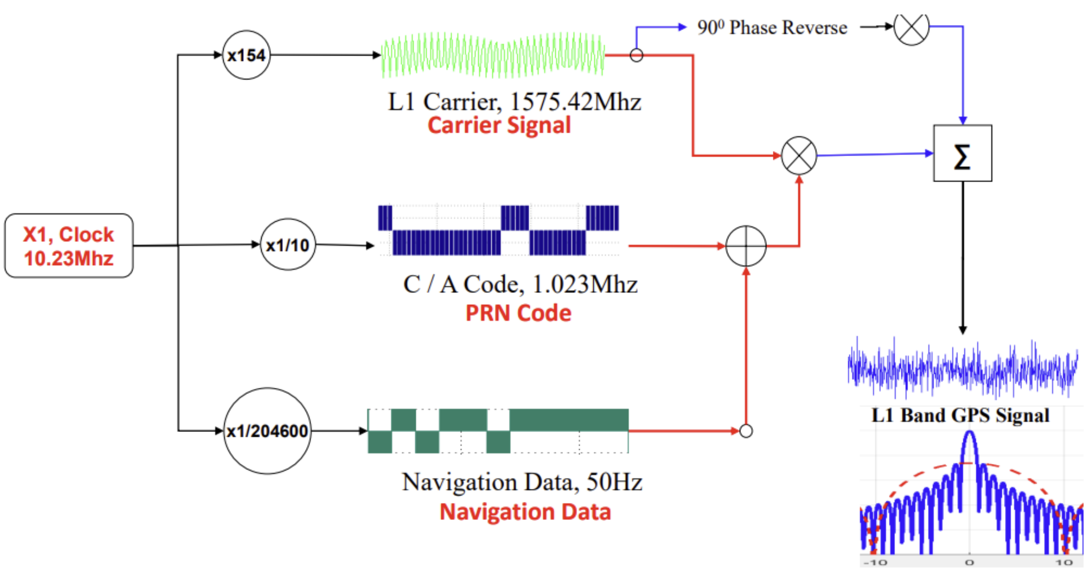
공식 메시지 배치와 본 시험 GNSS RAW Custom Message 위치

MSG ID	Message Title	Subframe	LNIS 1.0	시험 적용
G2	Health and Safety	SB2	Yes	참조
G4	Sorbit Ephemeris & Clock	SB2	Yes	AFS 필수정보
G5	MOrbit Almanac	SB3/SB4	Yes	선택
G8	Time of Transmission	SB2	Yes	AFS 시간
G12	Observation	SB3	Yes	미사용
Custom	LNSP-defined Custom Message	SB4	Allowed	GNSS RAW

7. GNSS 송신 데이터

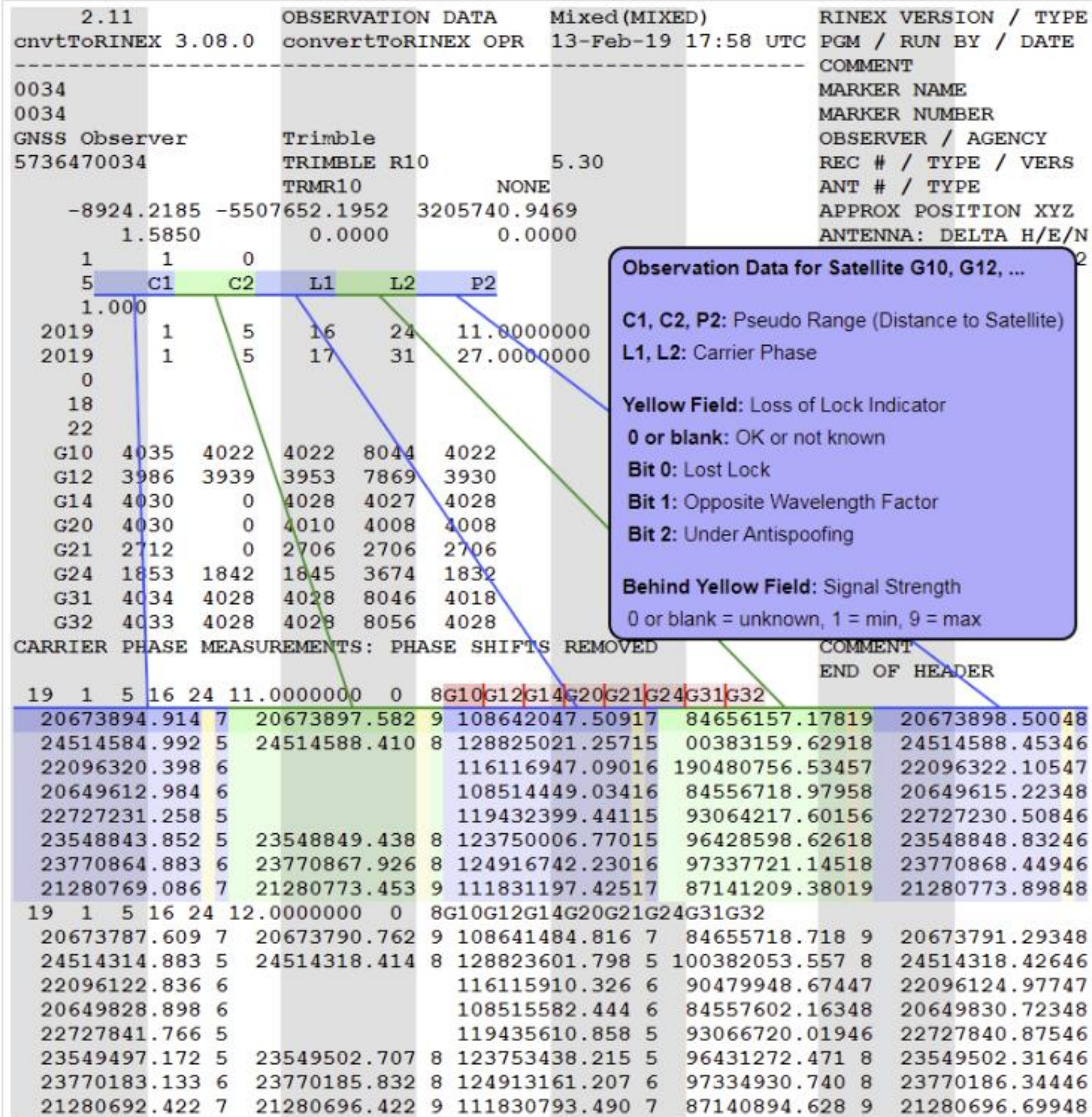


120 minutes (maximum value)
Message 1
...
Message i
...
Message N
Navigation Message
12 sec
8 bits
6 bits
6 bits
1 bit
17 bits
238 bits
24 bits
PREAMBLE
PRN
MSG TYPE ID
ALERT FLAG
MESSAGE TOW COUNT
...
CRC
Message
12 seconds



X1, Clock 10.23Mhz
x154
x1/10
x1/204600
L1 Carrier, 1575.42Mhz
Carrier Signal
90⁰ Phase Reverse
C / A Code, 1.023Mhz
PRN Code
Navigation Data, 50Hz
Navigation Data
Σ
L1 Band GPS Signal
-10
0
10

8. RINEX Observation data(GNSS Raw Data)



```
     2.11           OBSERVATION DATA    Mixed(MIXED)        RINEX VERSION / TYPE
cnvtToRINEX 3.08.0  convertToRINEX OPR  13-Feb-19 17:58 UTC PGM / RUN BY / DATE
------------------------------------------------------------COMMENT
0034                                                        MARKER NAME
0034                                                        MARKER NUMBER
GNSS Observer       Trimble                                 OBSERVER / AGENCY
5736470034          TRIMBLE R10         5.30                REC # / TYPE / VERS
                    TRMR10          NONE                    ANT # / TYPE
    -8924.2185 -5507652.1952  3205740.9469                  APPROX POSITION XYZ
        1.5850        0.0000        0.0000                  ANTENNA: DELTA H/E/N
     1     1     0                                                             2
     5    C1    C2    L1    L2    P2
     1.000
  2019     1     5    16    24   11.0000000
  2019     1     5    17    31   27.0000000
     0
    18
    22
   G10  4035  4022  4022  8044  4022
   G12  3986  3939  3953  7869  3930
   G14  4030     0  4028  4027  4028
   G20  4030     0  4010  4008  4008
   G21  2712     0  2706  2706  2706
   G24  1853  1842  1845  3674  1832
   G31  4034  4028  4028  8046  4018
   G32  4033  4028  4028  8056  4028
CARRIER PHASE MEASUREMENTS: PHASE SHIFTS REMOVED            COMMENT
                                                            END OF HEADER
 19  1  5 16 24 11.0000000  0  8G10G12G14G20G21G24G31G32
  20673894.914 7  20673897.582 9 108642047.50917  84656157.17819  20673898.50048
  24514584.992 5  24514588.410 8 128825021.25715  00383159.62918  24514588.45346
  22096320.398 6                 116116947.09016 190480756.53457  22096322.10547
  20649612.984 6                 108514449.03416  84556718.97958  20649615.22348
  22727231.258 5                 119432399.44115  93064217.60156  22727230.50846
  23548843.852 5  23548849.438 8 123750006.77015  96428598.62618  23548848.83246
  23770864.883 6  23770867.926 8 124916742.23016  97337721.14518  23770868.44946
  21280769.086 7  21280773.453 9 111831197.42517  87141209.38019  21280773.89848
 19  1  5 16 24 12.0000000  0  8G10G12G14G20G21G24G31G32
  20673787.609 7  20673790.762 9 108641484.816 7  84655718.718 9  20673791.29348
  24514314.883 5  24514318.414 8 128823601.798 5 100382053.557 8  24514318.42646
  22096122.836 6                 116115910.326 6  90479948.67447  22096124.97747
  20649828.898 6                 108515582.444 6  84557602.16348  20649830.72348
  22727841.766 5                 119435610.858 5  93066720.01946  22727840.87546
  23549497.172 5  23549502.707 8 123753438.215 5  96431272.471 8  23549502.31646
  23770183.133 6  23770185.832 8 124913161.207 6  97334930.740 8  23770186.34446
  21280692.422 7  21280696.422 9 111830793.490 7  87140894.628 9  21280696.69948
```


Observation Data

9. RINEX Navigation Message(GNSS Raw Data)

PRN

Navigation Data

```
     2.11           NAVIGATION DATA     GPS(GPS)            RINEX VERSION / TYPE
cnvtToRINEX 2.90.0  convertToRINEX OPR  05-Jul-17 03:38 UTC PGM / RUN BY / DATE
------------------------------------------------------------COMMENT
    0.8382D-08  0.2235D-07 -0.5960D-07 -0.1192D-06          ION ALPHA
    0.8602D+05  0.6554D+05 -0.1311D+06 -0.4588D+06          ION BETA
   -0.931322574615D-09-0.355271367880D-14   405504     1947 DELTA-UTC: A0,A1,T,W
    18                                                      LEAP SECONDS
                                                            END OF HEADER
32 17 05 01 00 00  0.0-0.400723423809D-03-0.110276232590D-10 0.000000000000D+00
    0.370000000000D+02-0.806250000000D+01 0.455840416154D-08-0.192420920137D+01
   -0.353902578354D-06 0.111064908560D-02 0.826455652714D-05 0.515371503258D+04
    0.864000000000D+05-0.782310962677D-07 0.675647076441D-01-0.838190317154D-07
    0.958529124300D+00 0.221156250000D+03-0.265074890978D+01-0.796390315710D-08
   -0.389659088008D-09 0.100000000000D+01 0.194700000000D+04 0.000000000000D+00
    0.240000000000D+01 0.000000000000D+00 0.465661287308D-09 0.370000000000D+02
    0.795120000000D+05 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
24 17 05 01 00 00  0.0-0.341213308275D-04-0.454747350886D-12 0.000000000000D+00
    0.100000000000D+02 0.787812500000D+02 0.459340561950D-08 0.167267059468D+01
    0.404566526413D-05 0.564297637902D-02 0.102464109659D-04 0.515370226479D+04
    0.864000000000D+05-0.782310962677D-07 0.108986675687D+01 0.484287738800D-07
    0.945651423640D+00 0.170906250000D+03 0.490563049326D+00-0.815641117584D-08
   -0.128933942045D-09 0.100000000000D+01 0.194700000000D+04 0.000000000000D+00
    0.240000000000D+01 0.000000000000D+00 0.279396772385D-08 0.100000000000D+02
    0.792180000000D+05 0.400000000000D+01 0.000000000000D+00 0.000000000000D+00
```

10. PVT Solve

① Observation
Pseudorange ρ
Doppler D
(C/N₀)

② Navigation Data
Ephemeris
Satellite Clock

③ 위성 위치·속도 계산
Ephemeris → Xs, Ys, Zs / Vs

④ 측정값 보정
Satellite Clock
전파시간 / 지구회전

⑤ PVT Solver
Pseudorange → Position
Doppler → Velocity
Clock Bias → Time

위성의 위치 + 각 위성까지의 거리를 → 수신기 위치

X, Y, Z + Receiver Clock Bias
→ 최소 위성 4개의 Pseudorange 필요

Position 계산
$\rho_i \approx$ 위성 i ↔ 수신기 거리
+ Receiver Clock Bias
4개 이상 위성 → 반복 최소제곱

최종 출력
Position : Latitude / Longitude / Height
Velocity : Vx / Vy / Vz
Time : GNSS Time 보정
= **PVT**

RINEX Observation + RINEX Navigation → PVT